

Reading a Coronary Artery's Shape

What we measure, what we dropped, and why

A plain-math companion to the CAD-prediction feature-selection reports
No background in differential geometry assumed.

1 What this document is

Across several rounds of literature research, this project narrowed down which shape-based measurements of a coronary artery are worth feeding into a coronary-artery-disease (CAD) prediction model, and which ones look plausible but don't hold up. This document collects the *final, locked-in* choices in one place, and explains the underlying math from scratch, so that the choices make sense even to someone seeing terms like “curvature” or “torsion” for the first time.

In plain words

Nothing here is a new claim. Every decision below was reached in the four research reports listed in Section 8, each of which cites the primary literature behind it. This document is the short, math-first version of those reports.

2 Two ways of looking at the same vessel

The project compares two models side by side, and the whole feature-selection exercise is really about feeding each one the right kind of input.

Model 1 is an ordinary logistic regression. It gets a short list of *numbers* that summarize each vessel, such as one average curvature value per vessel, plus age and sex. Think of Model 1 as being told “*on average, how winding is this hiking trail?*”

Model 2 is a small neural network. Instead of one summary number per vessel, it reads an *ordered list* of local measurements, one value for every point along the vessel's centerline, from the start of the vessel to its end. Think of Model 2 as being handed the trail's full *elevation profile*, step by step, rather than a single average.

Model 1: logistic regression	Model 2: small network
A handful of scalar numbers per patient	An ordered vector per vessel, one number per point
“the average bend of this vessel”	“the bend at every step along this vessel”
Age, sex, and a few geometric summaries	Age, sex, plus a per-point profile per vessel

Everything from here on is about deciding, for each candidate measurement: does it belong in Model 1's short list, in Model 2's per-point profile, in both, or in neither.

3 The building blocks: a plain-math tour

Before the decisions make sense, six ideas need defining. None of them require more than a ruler and a protractor to understand, even though the formulas look formal on paper.

3.1 The centerline: turning a vessel into a list of points

Every measurement below starts the same way. Software traces a thin line down the exact middle of the vessel, called the **centerline**, then places points along it at even spacing (here, every 0.5 mm). Walking along these points *in order*, from the vessel's start to its end, is what turns a 3D tube into something both a spreadsheet and a neural network can read: an ordered sequence of numbers. Model 2's "local vector" is exactly this walk, with one measurement recorded at each step.

3.2 Curvature: how sharply the path bends

Curvature answers one question at each point: *if you were driving along this vessel, how hard would you have to turn the wheel right here?* A straight stretch has zero curvature. A tight bend has high curvature.

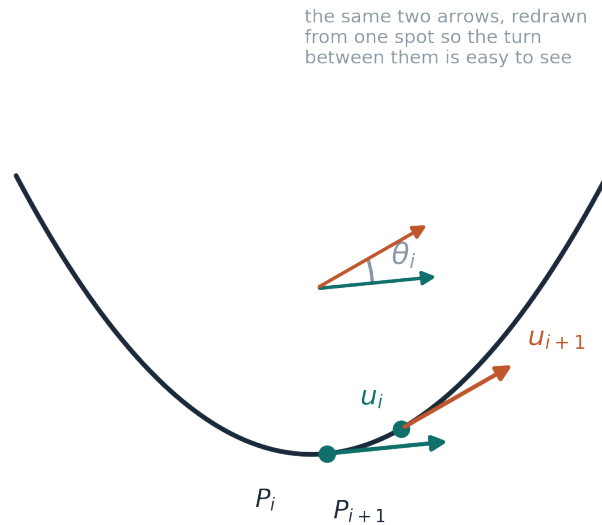


Figure 1: At each point P_i , the direction you're currently heading is a short arrow u_i (a *unit tangent vector*). Comparing the direction at one point to the direction at the next point gives a turning angle θ_i . A vessel that barely changes direction has a small θ_i everywhere; a winding vessel has large θ_i values throughout.

Averaging that turning angle over the whole vessel gives **mean absolute curvature**:

$$\kappa_a = \frac{1}{L} \sum_j \theta_j, \quad \theta_j = \arccos(u_j \cdot u_{j+1})$$

where L is the vessel's total length and $u_j \cdot u_{j+1}$ is the standard dot product between two direction arrows (which is how you compute the angle between two directions). There is also a close cousin, **RMS**

curvature, which combines the same per-point turning angles using a root-mean-square instead of a plain average: a slightly different way of averaging the same underlying signal.

3.3 Torsion: bending *out of the plane*

Curvature alone only sees bending *within a single flat plane*, like a road drawn on a flat map. But a real artery can also twist like a corkscrew, lifting out of whatever plane it was bending in a moment ago. **Torsion** is the measurement built to catch exactly that.

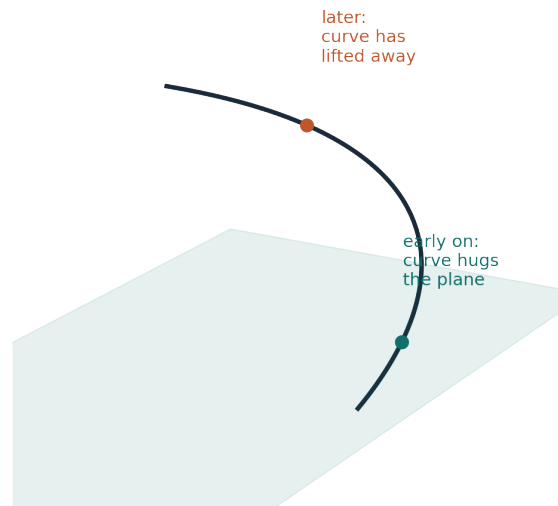


Figure 2: Early on, this curve bends inside one flat plane (shaded). Further along, it lifts away from that plane and starts bending in a new direction entirely. Torsion measures how fast the curve escapes its current bending plane; curvature alone cannot see this at all.

Mechanically, torsion is built the same way as curvature, an average turning angle, $\tau_a = \frac{1}{L} \sum_j \phi_j$, except ϕ_j tracks the turning of the *plane* itself (technically, the binormal-normal frame from the Frenet–Serret construction used throughout differential geometry) rather than the turning of the direction arrow u_j . The important practical fact is not the formula, it is what computing it *requires*: curvature needs two neighboring directions, but torsion needs to track how those directions’ supporting plane changes, which means it depends on a finer level of detail in the traced centerline. That finer dependency is exactly what makes it sensitive to small tracing noise, a point that matters a great deal in Section 4.

3.4 Bifurcation angle: how wide a fork in the road is

At a **bifurcation**, a vessel splits into two daughter branches. Each daughter heads off in its own direction, d_1 and d_2 . The **bifurcation angle** is simply the angle between those two directions:

$$\theta_{bif} = \arccos(d_1 \cdot d_2)$$

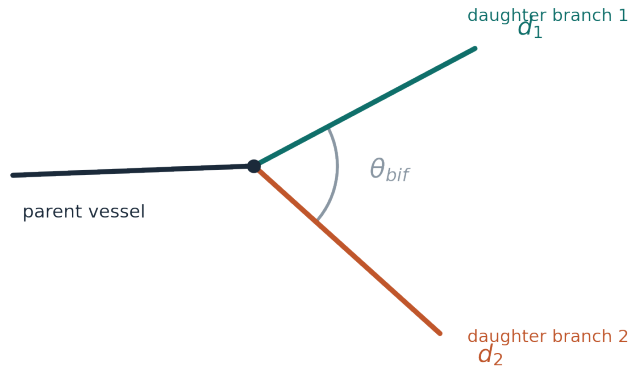


Figure 3: The parent vessel splits into two daughters. The angle between the two daughters' directions, θ_{bif} , is the bifurcation angle. Unlike curvature, this is a single number attached to one specific location, not something computed at every point along a vessel.

3.5 Diameter: how wide the tube is

The simplest measurement of all: the vessel's width at a given point. Diameter is not constant along a vessel, it tapers gradually from the origin toward the tip, and it steps down slightly, and normally, wherever a branch splits off and carries some of the flow away.

3.6 Coronary dominance: which artery reaches the back of the heart

This one is not a shape measurement, it is closer to a category, like sex. On the back/underside of the heart, either the right coronary artery (RCA) or the left circumflex artery (LCx) supplies the final stretch of vessel (the posterior descending artery). Whichever one does is called **dominant**. Roughly 80–85% of people are right-dominant, a smaller group left-dominant, and a few are co-dominant. Dominance does not measure disease, but it does change how long or short some vessels naturally are, which matters when a model needs to tell “this vessel looks different because of anatomy” apart from “this vessel looks different because of disease.”

3.7 Horton–Strahler ratio: counting branch levels

A classical idea, originally from mapping river networks, later reused for tree-shaped biological structures. Label the smallest end-twigs of a branching tree “order 1.” Whenever two order-1 branches merge, the result becomes “order 2,” and so on, all the way up to the main trunk. The **Horton–Strahler ratio** compares how many segments exist at one order versus the next:

$$R_B \approx \frac{N_i}{N_{i+1}}$$

For this ratio to mean anything stable, the tree needs *several* consecutive orders to compare, five or more in every study that reports it as reliable. A tree with only two or three visible levels of branching gives a

ratio built from very small integers (three segments over one, say), which swings wildly from patient to patient just from counting noise, not from any real anatomical difference.

4 The seven decisions, at a glance

With the vocabulary in place, here is what the research concluded for each candidate measurement.

Kept: mean absolute curvature

Used as Model 2's one and only curvature channel, per point, on the RCA, LAD, and LCx. It is the single measurement with a direct, published link to a coronary hemodynamic outcome (low wall shear stress), and the one study that tested it head-to-head against RMS curvature recommended standardizing on it.

Dropped: RMS curvature

RMS curvature is built from the exact same per-point turning angles as mean absolute curvature, just averaged differently. The two carry substantially overlapping information about the same underlying bending. Keeping both would spend model capacity, and later sample-size budget, on near-duplicate information instead of on a genuinely new signal.

Kept: diameter

Added as a second per-point channel in Model 2, alongside curvature, on the RCA, LAD, and LCx. It is the one candidate with direct outcome evidence as an *independent* predictor (smaller diameter, reduced functional flow), not merely as a normalizing denominator for some other metric.

Kept: bifurcation angle

Included, but only as a **scalar** covariate (one or two numbers per vessel, at fixed proximal landmarks), never as a dense per-point channel. Every vessel's outcome evidence concentrates at one or two specific junctions, not along a running series of branch points, even on vessels like the LAD that branch often.

Dropped: torsion, in every form

Excluded from Model 2's profile *and* from Model 1's scalar list. Two independent problems compound here: torsion needs a finer level of tracing detail than curvature, which makes it noise-sensitive by construction, and no study anywhere links torsion, in scalar or per-point form, to a coronary disease outcome. A proposed mechanistic backstop (torsion drives helical blood flow, which shapes wall shear stress) was directly tested in a 2025 coronary-specific study and did not hold up after correcting for multiple comparisons.

Kept: coronary dominance

Included as a scalar, categorical covariate (right / left / co-dominant), in both models. Not because dominance itself is a settled, independent CAD risk factor, the direct evidence for that is contested, but because it is a genuine confound: it measurably changes vessel length and course on all three major vessels, and a model that doesn't know about it risks reading normal anatomical variation as disease.

Dropped: Horton–Strahler ratio

Excluded entirely. Clinical CT scans of the coronary tree only resolve about 2–4 branch generations per named vessel, well short of the five or more consecutive orders the ratio needs to be numerically stable. No study anywhere has computed it on a CT-scanned human coronary tree, and no study links it to coronary disease at all.

Feature	Decision	Form	One-line reason
Mean absolute curvature	Kept	Per-point (Model 2)	Best-tested single curvature signal
RMS curvature	Dropped	n/a	Duplicates mean absolute curvature
Diameter	Kept	Per-point (Model 2)	Direct, independent outcome link
Bifurcation angle	Kept	Scalar (both models)	Evidence sits at fixed landmarks only
Torsion	Dropped	n/a	Noise-sensitive; zero outcome evidence
Coronary dominance	Kept	Scalar (both models)	Confound on all three vessels
Horton–Strahler ratio	Dropped	n/a	Coronary CT can't resolve enough orders

5 Vessel by vessel

The rules above are not identical for every vessel; anatomy gets in the way in three specific places.

RCA. The reference case: diameter and curvature travel along the whole vessel as a per-point profile; angle and dominance enter as scalars.

LAD. Same recipe, with one wrinkle: the LAD gives off several diagonal branches, and the vessel's diameter takes a small, completely normal step down at every one of them. A per-point diameter reading needs to know this, or it will mistake ordinary branch takeoffs for narrowing caused by disease.

LCx. Same recipe again, with a caution rather than a wrinkle: the LCx is consistently the *most* curved of the three vessels, but two separate studies directly disagree on whether a tortuous LCx means more or less disease. Because curvature magnitude alone can distinguish the LCx from the other vessels, a shared curvature channel risks quietly learning “which vessel is this” instead of “how diseased is this vessel,” a risk that is sharper here than on the RCA or LAD.

Left main (LM). This is the one vessel where the answer is not “same recipe, different vessel,” it is a different representation altogether. The LM averages only about 10.5 mm in length, and can be as short as 2 mm. At this project's 0.5 mm point spacing, that is roughly 15 to 25 points end to end for a typical LM, and as few as 10 for a short one, nowhere near enough to treat as a meaningful profile. Every study found

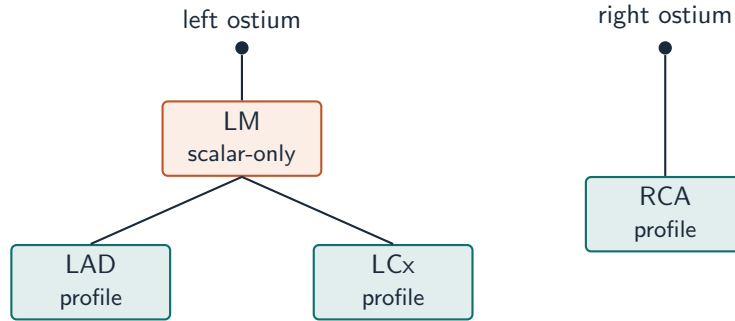


Figure 4: The coronary circulation is really two-to-three separate trees fed from two ostia. Three vessels (RCA, LAD, LCx) get a full per-point profile; the left main trunk (LM), where it is measured at all, is handled entirely differently (see below).

in this research that measures LM geometry at all reaches the same conclusion independently: it treats the LM as a small handful of scalar numbers, never as a point-by-point sequence. The recommended set, fused into the model alongside age and sex rather than read as a sequence, is:

- the LAD–LCx bifurcation angle at the LM's tip,
- Finet's ratio (how the two daughter diameters compare to the parent diameter, a classical branching-law prediction),
- the LM's diameter near its origin,
- the LM's diameter just before it splits,
- a yes/no flag for a third branch (a *ramus intermedius*), since a three-way split has a measurably different typical angle than a simple two-way split.

6 From one vessel to the whole tree

Combining four vessels into one model raises two questions that a single-vessel design never had to answer.

How does one network read four different vessels? The recommended design uses one *shared* reading module (the same small network weights) for the RCA, LAD, and LCx profiles, rather than three separate networks. To let the shared module still tell the vessels apart, each profile is tagged with a **vessel-identity** signal before it goes in, so the network can adjust its behavior depending on which vessel it is currently reading. The LM's five scalar numbers, plus age, sex, the bifurcation angle(s), and dominance, are fused in separately, near the output, rather than passed through the sequence-reading part of the network at all.

Does adding vessels add data? This is the counter-intuitive part. A patient's RCA, LAD, and LCx are not independent draws, they are three correlated views of the *same* patient's overall vessel health. Treating “4 vessels × N patients” as “4N independent training examples” overstates how much data the model actually has, in the same way that letting a hospital scan appear in both the training set and the test set would. The safe accounting stays close to the number of *patients*, not the number of vessel readings, and any evaluation must keep every vessel from one patient entirely on one side of a train/test split, never spread across both.

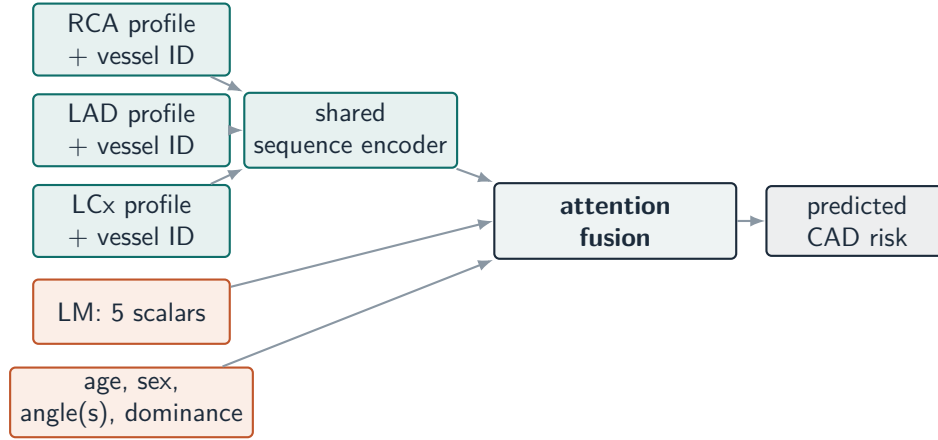


Figure 5: One shared encoder reads each of the three profile vessels, tagged with which vessel it is. Its outputs, the LM’s scalar features, and the remaining covariates are combined by an attention-based fusion step, which naturally copes with a patient missing one vessel’s clean extraction, before a final classifier produces one prediction per patient.

7 The final scalar checklist for Model 1

Applying every decision above, here is the concrete covariate list for the logistic regression, Model 1, across the whole tree:

#	Covariate	Domain	Parameters
1	Age	clinical	1
2	Sex	clinical	1
3	Mean absolute curvature, RCA	curvature	1
4	Mean absolute curvature, LAD	curvature	1
5	Mean absolute curvature, LCx	curvature	1
6	Bifurcation angle, LM take-off	angle	1
7–8	Coronary dominance (right / left / co-dominant)	anatomy	2

That is **eight predictor parameters** in total. This number is not arbitrary: classical sample-size guidance for logistic regression says the number of confirmed CAD cases needed per predictor roughly doubles, from about 10 to about 20, once a model grows past 5–6 predictors. Eight parameters is deliberately close to the edge of what a modestly sized cohort can support without the model’s apparent performance being an artifact of too little data for too many knobs. This is precisely why RMS curvature, torsion, and the Horton–Strahler ratio were pruned rather than added “for completeness”: each one would have pushed this list toward a regime the sample-size literature flags as actively unreliable.

Model 2, the small network, tolerates a longer scalar list more comfortably, because scalar covariates fused near its output cost only a small number of extra weights, unlike a per-point profile channel, which adds width all the way through the network. That said, no equivalent, numerically validated rule exists for neural networks the way it does for logistic regression, so this extra headroom should be treated as real,

but qualitative, not as license to add candidates without evidence.

8 Further reading

The full evidence trail, including every primary source, sample size, and effect size behind the decisions in this document, lives in four reports in this project's `reports/` folder:

1. *Tortuosity feature count for CAD prediction*: the original sample-size and redundancy arguments (events-per-variable, why mean absolute and RMS curvature overlap).
2. *Expanding local tortuosity vector features*: the RCA-specific case for adding diameter and keeping angle as a scalar.
3. *Tortuosity features across coronary tree*: extends the above to the LAD, LCx, and left main, and sets the whole-tree architecture.
4. *Scalar tortuosity metrics for CAD models*: the most recent round: the Horton–Strahler ratio, dominance, and the final decision to drop torsion entirely.

This document summarizes locked-in feature-engineering decisions for the CASNet/ImageCAS-X CAD-prediction thesis work. It is a plain-language companion, not a replacement for the cited reports' evidence and citations.